Special Tutorial Session for CAT-1 Preparations

Part – III (Updated)

1. In a cricket tournament, a batsman plays 150 balls. On each ball, he scores either 1 run (a single) or 6 runs (a six). If we pick a ball at random, the expected runs per ball is 2.5. How many singles and how many sixes did he hit?

2. At a cinema hall, customers can buy a combo that always includes: one snack, one drink, and one dessert. The choices available: Snack: Popcorn **or** Nachos (2 choices) Drink: Coke, Pepsi, or Sprite (3 choices), Dessert: Brownie **or** Ice-cream (2 choices). Determine the following:

i. How many different combos can be ordered?
ii. If every item is equally likely to be chosen, what is the probability that the combo has Popcorn with Coke**?**

3. The SSN Music Club (SMC) has 25 songs in its playlist, numbered 1 to 25. During an event, the DJ plays 4 random songs one after the other, without repeating any song**.** The rule of the game: If the 4 chosen songs are played in ascending order of their numbers (e.g., 3, 7, 14, 20), then the audience wins a surprise pizza party**.** What is the probability that the audience wins the pizza party?

4. Two rival hacker groups, **RedCode** and **BlueShield**, are battling in a cybersecurity challenge.

- **RedCode** has **n exploits** (attacks) prepared. Each exploit is labeled with a number chosen from {1, 2, 3, …, k}. (Numbers may repeat.)
- **BlueShield** has **k defenses** prepared. Each defense is labeled with a number chosen from {1, 2, 3, …, n}. (Numbers may repeat.)

The hackers line up in two queues: one queue of RedCode (attacks), one queue of BlueShield (defenses).
We define a sequence s0, s1, s2, … where s0 = 0. Then:

- If sj ≤ 0, RedCode plays first: take the first exploit number x from the RedCode queue, remove it, and set
  sj+1= sj + x
- If sj > 0, BlueShield responds: take the first defense number y from the BlueShield queue, remove it, and set
  sj+1= sj - y

We repeat this process until one queue becomes empty.
Finally, we create a **histogram of the sequence values** (s0, s1, s2, …). Between what range (on the X-axis) can the histogram exist?

5. Imagine you are using Bluetooth earphones to listen to music during a bus ride. The signal travels in two stages**:**

- **Mobile → Bluetooth transmitter**
- **Bluetooth transmitter → Earphones**

Each stage can introduce noise (distortion), flipping the bits (0 or 1) being transmitted.

- If a **0** is sent:
  - Stage-1 (mobile → transmitter) delivers correctly with probability **0.95**.
  - Stage-2 (transmitter → earphones) delivers correctly with probability **0.9**.
- If a **1** is sent:
  - Stage-1 delivers correctly with probability **0.9**.
  - Stage-2 delivers correctly with probability **0.85**.

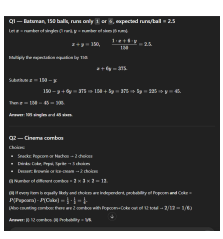
Assume **0 and 1 are equally likely** (music is compressed binary data).

a) Compute the probability that the earphones **finally receive a 0 when 0 was sent.**
b) Compute the probability that the earphones **finally receive a 1 when 1 was sent**.
c) Using these, calculate the **entropy** of the received signal.

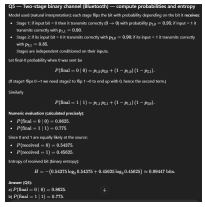
6. Six students across campus each write down the last digit of their mobile phone number (digits 0–9). We are only interested in these last digits **mod 5** (that is, the remainder when dividing by 5). So each student's digit will fall into one of the five groups: {0, 1, 2, 3, 4}. If you check all possible pairs among the six students, what is the probability that the difference of their two mobile numbers is divisible by 5 (equivalently, both students have the same remainder mod 5)? Also, compute the entropy of this experiment.

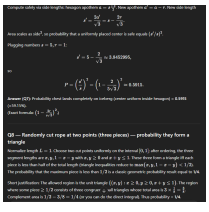
7. An adventurous explorer is trying to drop a treasure chest onto a floating hexagonal iceberg in the Arctic. The iceberg has a regular hexagon shape with a side of 5 meters. The treasure chest has a circular base of radius 1 meter. Due to wind and waves, the explorer can only aim roughly at the iceberg. What is the probability that the treasure chest lands completely on the iceberg without falling into the freezing water?

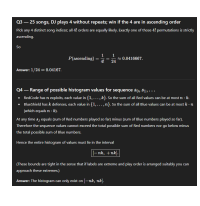
8. A sailor has a rope of length *L*. He randomly cuts the rope at two points along its length, producing three pieces. The sailor wants to use these pieces to tie **a** triangular sling for lifting cargo. What is the probability that the three resulting pieces can form a triangle?

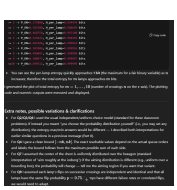


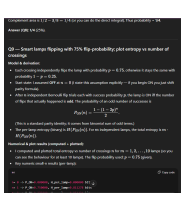
9. Imagine a corridor with a row of smart lamps**.** Every time a person passes under a lamp, the lamp flips its state: **ON → OFF** or **OFF → ON**. However, due to faulty wiring, with 25%











Setup / model (restating cleanly)Two-stage channel: Mobile → Transmitter → Earphones.Stage 1 correctness probabilities:If input = 0, stage-1 transmits correctly (0→0) with • 1,0=0.95p1,0 =0.95. So stage-1 flips 0→1 with probability 1−□1,0=0.051−p1,0 =0.05.If input = 1, stage-1 transmits correctly (1→1) with • 1,1=0.90p1,1 =0.90. So it flips 1→0 with probability 0.100.10.Stage 2 correctness probabilities (conditioned on what stage-2 receives):If its input = 0, it transmits correctly with • 2,0=0.90p2,0 =0.90 (so it flips 0→1 with probability 0.100.10).If its input = 1, it transmits correctly with • 2,1=0.85p2,1 =0.85 (so it flips 1→0 with probability 0.150.15).Input bits are equally likely: • (• =0)=• (• =1)=0.5P(X=0)=P(X=1)=0.5.Stages act independently given their inputs. We want: (a) • (final=0• Ł Ł =0)P(final=0• X=0), (b) • (final=1• Ł Ł =1)P(final=1• X=1), (c) entropy of the received bit • Y.(a) • (final=0• Ł Ł =0)P(final=0• X=0) — detailed path breakdownIf the source sends 0, there are two disjoint ways the earphones can finally get 0:Stage-1 transmits correctly (0→0) and stage-2 also transmits correctly (0→0).Probability = • 1,0 ⋅ ⊠2,0p1,0 ⋅ p2,0 .Stage-1 flips the 0 to a 1 (probability 1−□1,01−p1,0), but then stage-2 flips that 1 back to 0 (probability 1−□2,11−p2,1).Probability = (1−□1,0) ⋅ (1−□2,1)(1−p1,0) ⋅ (1−p2,1).These are mutually exclusive events, so add them:• (final=0• Ł Ł =0)=• 1,0• 2,0+(1−□1,0)(1−□2,1).P(final=0• X=0)=p1,0 p2,0 +(1−p1,0)(1−p2,1).Now plug numbers (do arithmetic carefully):• 1,0• 2,0=0.95×0.90=0.855p1,0 p2,0 =0.95×0.90=0.855.(1−□1,0)(1−□2,1)=0.05×0.15=0.0075(1−p1,0)(1−p2,1)=0.05×0.15=0.0075.Sum: 0.855+0.0075=0.8625.0.855+0.0075=0.8625.So• (final=0• Ł Ł =0)=0.8625.P(final=0• X=0)=0.8625.You can also check the complementary probability (error when 0 was sent):• (final=1• Ł Ł =0)=1−0.8625=0.1375,P(final=1• X=0)=1−0.8625=0.1375,which equals • 1,0(1−□2,0)+(1−□1,0)• 2,1p1,0 (1−p2,0)+(1−p1,0)p2,1 (the two ways to end as 1).(b) • (final=1• Ł Ł =1)P(final=1• X=1) — symmetric reasoningIf the source sends 1, the earphones get 1 in two ways:Stage-1 transmits correctly (1→1) and stage-2 transmits correctly (1→1): probability • 1,1• 2,1p1,1 p2,1 .Stage-1 flips 1→0 (prob 1−□1,11−p1,1), then stage-2 flips 0→1 (prob 1−□2,01−p2,0). Probability = (1−□1,1)(1−□2,0)(1−p1,1)(1−p2,0).So• (final=1• Ł Ł =1)=• 1,1• 2,1+(1−□1,1)(1−□2,0).P(final=1• X=1)=p1,1 p2,1 +(1−p1,1)(1−p2,0).Numbers:• 1,1• 2,1=0.90×0.85=0.765.p1,1 p2,1 =0.90×0.85=0.765.(1−□1,1)(1−□2,0)=0.10×0.10=0.01.(1−p1,1)(1−p2,0)=0.10×0.10=0.01.Sum: 0.765+0.01=0.775.0.765+0.01=0.775.So• (final=1• Ł Ł =1)=0.775.P(final=1• X=1)=0.775.Complementary error when 1 was sent:• (final=0• Ł Ł =1)=1−0.775=0.225.P(final=0• X=1)=1−0.775=0.225.Channel transition matrix (compact view)Write • (• Ł •)P(Y• X) as a 2×2 matrix (rows = X=0,1; columns = Y=0,1): • =0 • =1• =0 0.8625 0.1375• =1 0.2250 0.7750X=0X=1 Y=00.86250.2250 Y=10.13750.7750 This shows the channel is asymmetric (error probability depends on whether a 0 or 1 was sent). Error rates:Error given 0: 0.13750.1375 (13.75%).Error given 1: 0.2250.225 (22.5%).Average bit-error rate (with equiprobable inputs) = (0.1375+0.225)/2=0.18125(0.1375+0.225)/2=0.18125 ≈ 18.125%.(c) Distribution of received bit • Y and its entropySince inputs are equally likely • (• =0)=• (• =1)=0.5P(X=0)=P(X=1)=0.5, the marginal probability that the earphones see a 0 is• (• =0)=0.5 ⋅ ⊠(• =0• Ł Ł =0)+0.5 ⋅ ⊠(• =0• Ł Ł =1)=0.5(0.8625)+0.5(0.225)=0.54375.P(Y=0)=0.5 ⋅ P(Y=0• X=0)+0.5 ⋅ P(Y=0• X=1)=0.5(0.8625)+0.5(0.225)=0.54375.Thus • (• =1)=1−0.54375=0.45625.P(Y=1)=1−0.54375=0.45625.Shannon entropy (binary) of • Y:• (•)=−(0.54375log• 20.54375+0.45625log• 20.45625).H(Y)=−(0.54375log2 0.54375+0.45625log2 0.45625).Evaluating this numerically gives• (•)≈0.99447 bits.H(Y)≈0.99447 bits. Interpretation: The received bit • Y is almost maximally uncertain (a fair coin has • =1H=1 bit). Because the average bias is small (• (• =0)=0.54375P(Y=0)=0.54375 not far from 0.5), • Y's entropy is close to 1 bit.Extra information-theoretic quantities (useful perspective)Conditional entropy • (• Ł •)H(Y• X) — the average uncertainty about • Y given we know • X (i.e., how noisy the channel itself is):• (• Ł •)=12• (• (• =0• Ł Ł =0))+12• (• (• =1• Ł Ł =1)),H(Y• X)=12H(P(Y=0• X=0))+21 H(P(Y=1• X=1)),where • (•)=−□log• 2• −(1−□)log• 2(1−□)H(p)=−plog2 p−(1−p)log2 (1−p).Calculate:• (• Ł • =0)=• (0.8625)≈0.57765H(Y• X=0)=H(0.8625)≈0.57765 bits.• (• Ł • =1)=• (0.775)≈0.76919H(Y• X=1)=H(0.775)≈0.76919 bits.So• (• Ł •)≈0.5(0.57765+0.76919)=0.67342 bits.H(Y• X)≈0.5(0.57765+0.76919)=0.67342 bits.Mutual information • (• ;•)=• (•)−□(• Ł •)I(X;Y)=H(Y)−H(Y• X) measures how many bits of information about the input • X are carried by the received • Y:• (• ;•)≈0.99447−0.67342=0.32105 bits.I(X;Y)≈0.99447−0.67342=0.32105 bits.Interpretation: With this noisy two-stage channel and uniform inputs, each transmitted bit conveys on average only about 0.32 bits of information to the earphones. The channel is fairly noisy in the information sense, even though • Y itself looks nearly random (high entropy); a large portion of • Y's uncertainty is independent of the input (that's • (• Ł •)H(Y• X)).Intuition & takeawaysThe final-correct probability for a bit is not simply the product of the two “correct” probabilities because double errors can cancel: e.g., 0→1 at stage-1 then 1→0 at stage-2 returns the original 0. That’s why we add the two terms • 1,• Ł Ł 2,• p1,b p2,b and (1−□1,•)(1−□2,1−□)(1−p1,b)(1−p2,1−b